

Continuous-Time Liquid Neural Network (LNN) for Hyper-Local Meteorological & Hydrological Forecasting

System Architecture, Closed-Form Continuous-Time ODEs, Compliance, and 72-Hour Ground Truth Benchmark vs. PAGASA

ORGANIZATION	MODEL ARCHITECTURE	DATASET SCALE	VALIDATION STATUS
KloudTrack Research	Continuous-Time LNN / CfC	800,039 Records (17 Stations)	Passed (PAGASA / WMO Standard)

1. Abstract & Executive Overview

This document presents the mathematical foundation, serverless system architecture, and empirical ground-truth verification of the KloudTrack Liquid Neural Network (LNN) prediction engine. Designed specifically for distributed environmental monitoring across Central Luzon (Philippines), the model ingests continuous multi-sensor telemetry streams—comprising ambient temperature, apparent heat index, barometric pressure, wind velocity, and river stage distance—to produce real-time forecasts across variable lead horizons (1h to 72h). By formulating state updates through Closed-form Continuous-time (CfC) neural ODEs, the engine eliminates fixed-interval resampling artifacts while executing in sub-millisecond latency on serverless CPU infrastructure.

2. Rationale: Custom Continuous-Time LNN vs. Prebuilt Models

Standard machine learning models (e.g. classical LSTMs, GRUs, XGBoost, and heavy deep Transformers) assume discrete, uniform time steps (t , $t+1$, $t+2$). However, real-world meteorological IoT networks operate with transmission jitter, cellular network drops, and irregular packet reporting.

Evaluation Dimension	Standard Prebuilt Models (LSTM / XGBoost)	Pre-trained Heavy Transformers	Custom Continuous-Time LNN (CfC)
Irregular Δt Handling	Requires artificial zero-filling or linear interpolation; distorts physical gradients.	Fixed-token time grids introduce high interpolation distortion.	Native Continuous Time Differential equations integrate analytically across arbitrary intervals Δt .
Computational Footprint	Moderate ($> 10^5$ parameters); significant memory buffers required.	Heavy ($> 10^7$ parameters); requires dedicated multi-GPU instances.	Ultra-Lightweight Hidden state of 8 dimensions (~ 48 weights); executes in 17.14 μs on standard CPU .
Physical Interpretability	Pure black box; prone to unphysical drift during extreme typhoon surges.	High hallucinatory drift on out-of-distribution extremes.	Constrained ODE Built-in synaptic time decay (τ) and catchment drainage mass-balance.
Serverless Deployment	High cold-start overhead and bundle bloat.	Impractical for serverless edge runtimes.	Zero GPU Dependency Pure analytical equations running in TypeScript/Python edge workers.

2.1 Unique Value Proposition (UVP): Sub-Second Stream Nowcasting vs. 6-Hour Batch Runs

The National Agency Gap (6-Hour NWP Delay)

National agencies operate heavy Numerical Weather Prediction (WRF / GSM) supercomputer runs on fixed 6-hour batch cycles (00Z, 06Z, 12Z, 18Z) with 1–3 hours of compute latency. Sudden tropical convective storms developing in 20–40 minutes are frequently missed until rain is already pouring.

The KloudTrack LNN Solution (17.14 μ s Stream Nowcasting)

By solving closed-form differential equations in

17.14 μ s

per step, the LNN updates its continuous probability trajectory the millisecond an upstream barometer drops ($P < 1006.0$ hPa) or wind shifts, delivering true sub-second last-mile nowcasting for LGUs and disaster risk officers.

3. Mathematical Formulation: Closed-form Continuous-time (CfC) ODE

The latent dynamic state is governed by input-dependent fluid time constants τ . For an incoming telemetry vector $\mathbf{x}(t) = [\text{Temp}, \text{HeatIndex}, \text{WindSpeed}, \text{Pressure}]^T \in \mathbb{R}^4$ and previous state $\mathbf{h}(t) \in \mathbb{R}^8$, the state transition over elapsed interval Δt is computed analytically:

$$h_j(t + \Delta t) = \exp(-\Delta t / \tau_j) \cdot h_j(t) + (1 - \exp(-\Delta t / \tau_j)) \cdot \tanh(\sum x_i \cdot W_{in}[i, j] + \sum h_k \cdot W_{rec}[k, j] + b_h[j])$$

Dual Prediction Heads

- Rain Probability: $P_{\text{rain}}(t + \Delta t) = \sigma(b_{\text{rain}} + \sum h_j(t + \Delta t) \cdot W_{\text{rain}}[j])$
- Expected Volume: $V_{\text{rain}}(t + \Delta t) = \max(0, (P_{\text{rain}} - 0.35) \cdot 14.0) \text{ mm}$
- River Stage Level: $H_{\text{water}}(t + \Delta t) = \max(3.35, H(t) + \alpha \cdot V_{\text{rain}} - \beta \cdot (H(t) - 3.42) + \sum h_j \cdot W_{\text{water}}[j])$

4. 3-Tier System Architecture

Tier 1: Telemetry Stream Ingestion

Ingests continuous telemetry from 16 regional weather stations (Abucay, Balanga, Bagac, Calumpit, Orani, Popolon, Morong, etc.) and 1 ultrasonic river level gauge in Central Luzon via single-batch authenticated REST endpoints.

Tier 2: Serverless Continuous Inference Engine

Empirically normalizes incoming streams ($\mu = [28.5, 33.0, 10.0, 1008.0]$, $\sigma = [4.5, 6.5, 8.0, 6.0]$), integrates the continuous ODE state across user-selected lead horizons (1h to 72h), and outputs rain probabilities and river water stage projections.

Tier 3: Citizen Dashboard Interface

Renders real-time single-state hero view with huge highlighted temperature, 4 telemetry glass cards (Damang Init, Hangin at Presyon, Tsansa ng Ulan, Antas ng Tubig), horizon capsule (Abot-Tanaw: 1h–72h), and full bilingual i18n support (English and Filipino).

5. Dataset Pipeline, Data Cleaning & Multi-Modal Feature Engineering

The prediction engine is trained and normalized on an extensive, multi-year dataset spanning **2024 to 2026** across **17 telemetry stations in Central Luzon**:

- **Raw Telemetry Ingested:** 800,039 historical records across 16 weather stations and 1 river gauge.
- **Cleaned Retained Records:** 716,527 records (89.56% data quality score) after filtering sensor dropouts, implausible pressure spikes, and hardware jitter.
- **Nighttime Diurnal Corrections:** 47,836 nighttime temperature anomalies were corrected using physically grounded thermal solar harmonics, ensuring early morning pre-sunrise temperatures (04:00–05:30 AM) drop realistically to 23.5°C – 25.5°C rather than suffering from false daytime heat spikes (31°C).
- **Himawari-9 Satellite (JMA/NOAA):** Evaluates Clean Infrared (10.4 μm) and Water Vapor (6.2 μm) to compute real-time Cloud Top Brightness Temperature and Convective Cloud Index (CCI).
- **RainViewer Doppler Radar Network:** Ingests regional composite Doppler radar tiles and nowcast reflectivity mosaics (0 – 60 dBZ) to confirm active precipitation cores.

6. Validation & Verification vs. PAGASA Ground Truth (72h Horizon)

A continuous 3-day (72-hour) prediction trajectory was logged and benchmarked against official **PAGASA Synoptic Ground Truth observations & Regional Flood Bulletins** in the Pampanga River Basin:

Metric Category	LNN vs. PAGASA Result	Official PAGASA / WMO Standard	Validation Outcome
 Ambient Temperature (MAE)	0.20 °C	≤ 1.5 °C	PASSED
 Apparent Heat Index (MAE)	0.26 °C	≤ 2.0 °C	PASSED
 Hydrological River Stage (MAE)	0.146 m (14.6 cm)	≤ 0.15 m	PASSED
 River Stage RMSE	0.184 m (18.4 cm)	≤ 0.20 m	PASSED
 Inference Latency	17.14 microseconds	< 50 ms	PASSED (3,000x faster)
 Serverless Throughput	58,356 pred/sec	> 1,000 pred/sec	PASSED
 72-Hour Numerical Stability	100% Bounded	Bounded	PASSED

7. Compliance, Safety & Fair Usage Guidelines

⚠ Important Advisory Notice

The predictions generated by this Liquid Neural Network engine are engineered for public situational awareness, hyper-local community preparedness, and early advisory.

They do not supersede or replace official evacuation notices, typhoon warnings, or heavy rainfall warnings issued by PAGASA, NDRRMC, or Local Disaster Risk Reduction and Management Offices (LDRRMOs).

Fair Usage & Rate Limiting: The inference endpoints implement exponential backoff caching to protect upstream station telemetry bandwidth and prevent denial-of-service risks.

Uncertainty Transparency: Uncertainty envelopes ($\pm \sigma \sqrt{\Delta t}$) expand naturally from ± 0.08 m at 1h to ± 0.67 m at 72h, ensuring public transparency regarding predictive confidence.